

Term	Definition
Agile methodology	A set of principles for software development under which requirements and solutions evolve through collaborative effort.
Algorithm	A set of steps designed to solve a problem or perform a task. Used in programming to create software functions.
AI, Artificial Intelligence	Artificial Intelligence is the simulation of human intelligence in machines. Applied in speech recognition, robotics, and personal assistants.
API	Application Programming Interface. It's a set of rules and protocols that allow software applications to communicate and exchange data.
AR, Augmented reality	Augmented Reality is an interactive experience where digital information is overlaid onto the real world. Used in gaming and education.
Big data	Extremely large data sets analysed computationally to reveal patterns, trends, and associations. Used in marketing and research.
Binary	A base-2 number system that uses two digits, 0 and 1. Fundamental for computer data representation and operations.
Blockchain	A decentralised ledger of all transactions across a network. Used in cryptocurrencies and secure financial transactions.
Bug	An error or flaw in software that causes incorrect or unexpected results. Fixed through a process called debugging.
Cloud computing	Using remote servers hosted on the internet to store, manage, and process data. Provides flexible IT resources.
Computer vision	An AI technology that enables computers to interpret and make decisions based on visual data.
CDN	Content Delivery Network is a system of distributed servers that deliver web content to a user based on geographic locations.
CI, Continuous Integration	Continuous Integration is the practice of automating the integration of code changes from multiple contributors into a single software project.
Cybersecurity	The practice of protecting systems and networks from digital attacks. Essential for securing sensitive information.
Data	Facts and statistics collected for reference or analysis. Primarily used to base decisions or calculations on.
Data breach	An incident in which sensitive, protected, or confidential data is accessed or disclosed in an unauthorised way.
Data governance	The management of data's availability, usability, integrity, and security.
Data warehouse	A system used for reporting and data analysis, storing large quantities of historical data.
Database	An organised collection of data managed by a database management system (DBMS). Used for storing and retrieving large amounts of information efficiently.
DevOps	A culture and set of processes that bring development and operations teams together to complete software development.
Domain name	The address of a website on the internet. Used to locate websites and resources online.
DNS	Domain Name System is the hierarchical and decentralised naming system for devices connected to the internet or a private network.
Encryption	The process of converting data into a code to prevent unauthorised access.
ER, Entity-Relationship	Entity-Relationship is a graphical representation of entities and their relationships to each other. Used in database modelling.
Firewall	A network security system that monitors and controls incoming and outgoing network traffic. Helps protect against cyber threats.
Function	A block of reusable code designed to perform a specific task. Simplifies programming by breaking tasks into smaller parts.

GB, Gigabyte	Gigabyte is a unit of data equal to approximately one billion bytes. Used to measure digital storage capacity.
Git	A distributed version control system for tracking changes in source code.
Graph	A data structure consisting of nodes connected by edges. Used to model relationships between entities.
Hardware	The physical components of a computer system. Includes parts like the CPU, monitor, keyboard, and hard drive.
Hyperlink	A link in a document that opens another document or resource when clicked. Fundamental for navigating the web.
Index	A data structure that improves the speed of data retrieval. Commonly used in databases.
IoT, Internet of Things	Internet of Things is a network of physical objects embedded with sensors and software that connect and exchange data.
IP, Internet Protocol	Internet Protocol is the protocol that governs the format of data sent over the internet or local network. Ensures data reaches its destination.
Java	A high-level programming language used for building applications. Widely used for web and mobile app development.
Kanban	A scheduling system for lean and just-in-time (JIT) production.
Library	A collection of precompiled routines or functions a program can use. Helps in code reuse and efficiency.
LMS, Learning Management System	Learning Management System is a software application that helps manage, deliver, and track educational courses or training programmes.
Loop	A sequence that repeats until a specified condition is met. Used in programming for tasks that require iteration.
Machine learning	A subset of AI that uses statistical techniques to give computers the ability to 'learn' from data.
Malware	Software that is specifically designed to disrupt, damage, or gain unauthorised access to a computer system.
NLP, Natural Language Processing	Natural Language Processing is a field of AI that enables computers to understand and process human language.
Network	A group of interconnected computers and devices. Allows for resource sharing and communication.
OAuth	An open standard for access delegation commonly used for token-based authentication.
OOP, Object-Oriented Programming	Object-Oriented Programming is a programming paradigm based on the concept of 'objects', which contain data and methods. Promotes code reuse and organisation.
Open source	Software with source code that anyone may inspect, modify, and enhance. Fosters collaborative software development.
Phishing	A cyberattack that uses disguised emails or messages as a weapon.
Pixel	The smallest unit of a digital image or display. Used to measure screen resolution.
Protocol	A set of rules governing data communication. Ensures proper data exchange between devices.
QA, Quality Assurance	Quality Assurance is a way of preventing mistakes or defects in manufactured products and avoiding problems when delivering solutions or services.
Queue	A linear data structure that follows the First In First Out (FIFO) principle. Used in task scheduling.
RAM, Random Access Memory	Random Access Memory is a type of computer memory that can be accessed randomly. Used for storing working data and machine code currently used.
Ransomware	A type of malware that threatens to publish the victim's data or perpetually block access to it unless a ransom is paid.

Repository	A central location where data is stored and managed. Crucial in version control systems for tracking code changes.
RPA, Robotic Process Automation	Robotic Process Automation is the use of software robots to automate high-volume, repetitive tasks.
Router	A device that forwards data packets between computer networks. Essential for internet communication.
Scrum	An agile framework for developing, delivering, and sustaining complex products. Uses iterative progress through sprints.
SSL, Secure Socket Layer	Secure Socket Layer is a standard technology for keeping an internet connection secure and safeguarding any sensitive data sent.
Server	A computer or program that provides services to other computers or clients on a network.
SOAP, Simple Object Access Protocol	Simple Object Access Protocol is a protocol for exchanging structured information in the implementation of web services.
SDLC, Software Development Lifecycle	Software Development Lifecycle is a framework defining tasks performed at each step in the software development process.
Source Code	The original code written by a programmer. It must be compiled or interpreted to run on a computer.
Sprint	A set period during which specific work must be completed and made ready for review.
SQL, Structured Query Language	Structured Query Language is a standard language for managing and manipulating databases.
Stack	A data structure that follows the Last In First Out (LIFO) principle. Used in various computing tasks, such as function call management.
Synchronous	Synchronous refers to events that happen at the same time or in a coordinated manner. In the context of technology and communication, it often describes processes or systems where operations occur at the exact same time, such as video calls or live streaming.
URL, Uniform Resource Locator	Uniform Resource Locator is the address of a specific webpage or file on the internet.
Usability testing	A technique used in user-centred interaction design to evaluate a product by testing it on users.
UAT, User Acceptance Testing	User Acceptance Testing is testing performed by the client to certify the system behaves as expected.
Variable	A storage location paired with an associated symbolic name that holds data. Can be changed during program execution.
Version control	A system that records changes to a file or set of files over time so that you can recall specific versions later.
VM, Virtual Machine	Virtual Machine is an emulation of a computer system. Allows multiple operating systems to run on a single physical machine.
VPN, Virtual Private Network	Virtual Private Network is a service that encrypts your internet connection and hides your IP address.
Virus	Malicious software designed to harm, exploit, or otherwise compromise computer systems.
WAN, Wide Area Network	Wide Area Network is a network that covers a broad area, such as a city, country, or global connections.
Web Browser	An application for accessing and viewing websites. Examples include Chrome, Firefox, and Safari.
Wi-Fi	A technology that allows electronic devices to connect to a wireless local area network (WLAN). Used for internet access.